A PROJECT REPORT ON

CUSTOMER SENTIMENT ANALYSIS FOR DEMAND FORECASTING OF ELECTRONIC DEVICES USING MACHINE LEARNING TECHNIQUES

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE OF

BACHELOR OF ENGINEERING (COMPUTER ENGINEERING)

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CERTIFICATE

This is to certify that the project report entitled

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ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on “CUSTOMER SENTIMENT ANALYSIS FOR DEMAND FORECASTING OF ELECTRONIC DEVICES USING MACHINE LEARNING TECHNIQUES”

We would like to thank our project guide Prof. Dr. B. A. Sonkamble for giving us all the help and guidance we needed. We are really grateful to him for his kind support and valuable suggestions.

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ABSTRACT

In today's rapidly evolving digital landscape, the surge of e-commerce platforms and the pervasive influence of social media have significantly reshaped consumer behavior. Customers now openly express their preferences, opinions, and experiences across various online platforms, providing a wealth of unstructured data that reflects real-time sentiment trends. These sentiments, if effectively analyzed, can serve as strong indicators of public perception toward products and influence decision-making in sectors such as marketing, logistics, and sales planning. However, traditional demand forecasting methods that rely solely on historical sales figures often fail to capture these dynamic shifts in consumer interest.

This paper presents the implementation of a sentiment-driven demand forecasting system that integrates natural language processing, deep learning models, and statistical forecasting techniques. The pipeline begins with web scraping of real-time user reviews and social media data from platforms such as Amazon using BeautifulSoup. Following data acquisition, sentiment analysis is performed using a multi-model approach that combines rule-based techniques (VADER, TextBlob), pre-trained transformers (DistilBERT), and a custom LSTM-based neural network trained on over 3.6 million product reviews. These models assign sentiment scores to the text, which are then used as key features in a demand prediction model. A Bayesian smoothing algorithm is applied to ensure statistical stability across regions with sparse data, and a pooling mechanism integrates outputs from multiple models to generate final demand estimations.

The results demonstrate that incorporating sentiment analysis significantly enhances forecasting accuracy, especially in dynamic markets where user perception evolves rapidly. The proposed system not only improves predictive performance but also enables more responsive business strategies. Future work may focus on expanding the linguistic scope of sentiment analysis to include non-English reviews, deploying real-time dashboards for insights, and exploring generative AI models to simulate hypothetical demand scenarios under varying sentiment conditions.

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LIST OF ABBREVATIONS

- **VADER** – Valence Aware Dictionary and sEntiment Reasoner
- **BERT** – Bidirectional Encoder Representations from Transformers
- **API** – Application Programming Interface
- **LSTM** – Long Short-Term Memory
- **CLI** – Command Line Interface
- **HTTP** – Hyper Text Transfer Protocol
- **CSV** – Comma-Separated Values
- **JSON** – JavaScript Object Notation

- **SQL** – Structured Query Language
- **OS** – Operating System
- **SDLC** – Software Development Life Cycle
- **UML** – Unified Modeling Language
- **NLP** – Natural Language Processing

LIST OF FIGURES

CHAPTER 1
INTRODUCTION

1.1 Overview :

With the explosion of digital commerce and widespread use of social media platforms, consumer behavior has become more dynamic and complex. Real-time opinions, feedback, and discussions now significantly influence product success. In this context, understanding and predicting customer sentiment becomes crucial for anticipating market trends. Traditional forecasting approaches often overlook these evolving online signals, making them less effective in fast-paced digital markets. Our project aims to bridge this gap by leveraging sentiment analysis to support more accurate and real-time demand forecasting.

1.2 Motivation :

Many businesses struggle to adapt quickly to fluctuations in public interest, especially when those changes are driven by social media trends or online reviews. Inspired by this, we were motivated to explore how modern machine learning and natural language processing (NLP) techniques can be harnessed to interpret online sentiment and link it to demand patterns. The motivation stems from a desire to create a system that not only forecasts demand more accurately but also provides insight into the emotional and contextual drivers behind consumer interest.

1.3 Problem Definition & Objectives :

Problem Definition: Traditional forecasting models are primarily based on historical data and fail to incorporate real-time consumer sentiment, making them rigid and reactive rather than adaptive.

Objectives:

- To develop a system that collects real-time reviews and opinions from e-commerce and social platforms.
- To analyze these texts using various sentiment analysis techniques.
- To integrate sentiment scores into a machine learning model that forecasts product demand trends.
- To evaluate the accuracy of demand predictions using real-world data.

1.4 Project Scope & Limitations :

This project focuses on electronic devices listed on Amazon. The scope includes building a robust scraping pipeline, applying multiple sentiment models, and using the results for demand forecasting. The geographical scope is limited to reviews and tweets from India, USA, and the UK. Limitations include the challenges of noisy data, differences in slang and regional language, and model generalizability across product categories. The system does not currently support multilingual analysis or real-time deployment at industrial scale.

1.5 Methodologies of Problem Solving :

To solve the identified problem, we implemented a modular pipeline involving three core components:

- Data Collection: Real-time scraping of reviews using BeautifulSoup.
- Sentiment Analysis: Multi-model approach involving VADER, TextBlob, DistilBERT, and a custom LSTM model.
- Forecasting: Aggregated sentiment scores are fed into a Bayesian inference-based model for demand prediction, with geographical weighting and smoothing applied for fairness and stability.

CHAPTER 2

LITERATURE SURVEY

1: A Survey on Sentiment Analysis Methods, Applications, and Challenges

(https://link.springer.com/article/10.1007/s10462-022-10144-1)

Authors : Zubair Baig, Imran Yaqoob, et al.

Publication: Artificial Intelligence Review

Year: 2022

Short Review: This paper provides a comprehensive survey of sentiment analysis techniques, their applications, and the challenges faced in this domain. It covers various methods, including machine learning and deep learning approaches, and discusses their effectiveness in analyzing sentiments from different data sources. The review highlights the importance of sentiment analysis in understanding public opinion and its applications in fields like marketing, finance, and social media monitoring.

2: Big Data Analytics: A Survey

(https://journalofbigdata.springeropen.com/articles/10.1186/s40537-015-0015-2)

Authors: Min Chen, Shiwen Mao, et al.

Publication: Journal of Big Data

Year: 2015

Short Review: This survey covers the landscape of big data analytics, including tools, techniques, and applications. It highlights the importance of big data in making informed decisions and optimizing operations. The paper discusses various big data processing frameworks like Hadoop and Spark and delves into the analytics techniques that can be used to extract valuable insights from large datasets.

3: Applications of Machine Learning Techniques in Business: A Review

(https://www.mdpi.com/2076-3417/13/7/4550)

Authors: A. S. Dey, B. Sobhan, et al.

Publication: Applied Sciences

Year: 2023

Short Review: The paper reviews various machine learning applications in business, focusing on areas such as marketing, finance, and supply chain management. It discusses how machine learning models can improve business processes and decision-making by analyzing large volumes of data and identifying patterns. The review provides insights into the use of regression, classification, clustering, and reinforcement learning techniques in different business scenarios.

4: Sentiment Analysis on Social Media Data: A Review
(https://www.mdpi.com/2673-4591/59/1/68)
Authors: K. Chaturvedi, R. Singhal
Publication: Proceedings
Year: 2021
Short Review: This review focuses on sentiment analysis techniques applied to social media data. It discusses different methods, including lexicon-based and machine learning-based approaches, and their effectiveness in extracting sentiments from platforms like Twitter and Facebook. The paper also addresses the challenges of sentiment analysis on social media, such as the handling of noisy data and the detection of sarcasm and irony.

5: Advanced Methods for Big Data Analytics and Data Mining
(https://www.mdpi.com/2076-3417/13/13/7599)
Authors: H. Wang, M. Chen, et al.
Publication: Applied Sciences
Year: 2023
Short Review: The paper discusses advanced methods in big data analytics and data mining, including algorithms and frameworks that enhance data processing and analysis. It covers techniques such as deep learning, ensemble methods, and evolutionary algorithms, and their applications in fields like healthcare, finance, and social media. The review emphasizes the importance of scalability and real-time processing in big data environments.

6: Sentiment Analysis Papers with Code
(https://paperswithcode.com/task/sentiment-analysis)
Authors: Various
Publication: Papers with Code

Year: Ongoing

Short Review: This resource provides a collection of sentiment analysis papers along with their corresponding code implementations. It covers various approaches, datasets, and benchmarks used for sentiment analysis. The repository includes implementations of state-of-the-art models and techniques, making it a valuable resource for researchers and practitioners looking to implement sentiment analysis in their projects.

7: Twitter as Research Data

(https://www.cambridge.org/core/journals/politics-and-the-life-sciences/article/twitter-as-research-data/6B31D18C5E2F9B8F9C0301BFB05F1C27)

Authors: W. T. Ventura, J. F. P. da Silva

Publication: Politics and the Life Sciences

Year: 2021

Short Review: This article discusses the use of Twitter data for research purposes, including the challenges and opportunities it presents. It highlights the potential of Twitter data in providing real-time insights into public opinion and behavior. The paper also addresses ethical considerations and data privacy issues associated with using social media data for research.

8: Information Systems in Big Data Era: A Survey

(https://www.mdpi.com/2078-2489/11/7/341)

Authors: D. G. Marjanovic, M. Minovic, et al.

Publication: Information

Year: 2020

Short Review: The survey explores the role of information systems in the big data era, focusing on how they support data-driven decision-making processes in various domains. It covers topics such as data integration, data quality management, and the use of information systems for big data analytics. The paper emphasizes the importance of interoperability and scalability in modern information systems.

9: Big Data and Predictive Analytics: What's Next?

(https://journalofbigdata.springeropen.com/articles/10.1186/s40537-019-0254-8)

Authors: S. L. Smith, A. H. Taylor

Publication: Journal of Big Data

Year: 2019

Short Review: This paper discusses the future directions of big data and predictive analytics, highlighting emerging trends and technologies that will shape the field. It covers advancements in machine learning, artificial intelligence, and data visualization techniques that are expected to enhance predictive analytics capabilities. The review also explores the challenges and opportunities associated with the integration of big data and predictive analytics.

CHAPTER 3
SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Assumptions & Dependencies :

- It is assumed that users will have access to the internet for web scraping of live reviews.
- The project depends on the availability of data from Amazon, which may be subject to access restrictions or API limits.
- It is also assumed that the required Python libraries (e.g., BeautifulSoup, Transformers, TensorFlow, Scikit-learn) are supported and compatible with the selected development environment.

3.2 Functional Requirements :

3.2.1 System Feature 1: Data Collection Module

- Scrapes product reviews from Amazon using BeautifulSoup.
- Preprocesses the collected data by removing noise, correcting formatting, and standardizing content.

3.2.2 System Feature 2: Sentiment Analysis Engine

- Applies VADER, TextBlob, DistilBERT, and a custom LSTM model to evaluate the sentiment polarity of the textual data.
- Aggregates sentiment scores using a pooling function for more robust sentiment representation.
- Assigns a normalized sentiment score for each review/tweet.

3.2.3 System Feature 3: Demand Forecasting Module

- Incorporates sentiment scores, historical trends, and seasonal data into a machine learning model (Bayesian Inference with decision tree ensemble).
- Outputs predicted demand values by region.
- Uses geographical sentiment smoothing and normalization for fairer unit distribution.

3.3 External Interface Requirements :

3.3.1 User Interfaces :

- Command-line interface (CLI) for executing different modules (scraping, sentiment analysis, forecasting).
- Optionally, graphical visualization of sentiment trends and demand forecasts via matplotlib/seaborn.

3.3.2 Hardware Interfaces :

- No special hardware interfaces are required beyond a standard computer system with internet access.

3.3.3 Software Interfaces :

- Amazon web pages for data input.
- Python interpreter for running scripts.
- Libraries including BeautifulSoup, NLTK, SpaCy, Transformers, TensorFlow, NumPy, Pandas, Scikit-learn, and Matplotlib.

3.3.4 Communication Interfaces :

- HTTP/HTTPS protocols for accessing web data.

3.4 Non-Functional Requirements :

3.4.1 Performance Requirements :

- The system must be capable of scraping, processing, and analyzing a minimum of 1,000 reviews/tweets within 10 minutes.
- Sentiment analysis should operate with at least 80% classification accuracy.

3.4.2 Safety Requirements :

- The system must not interact with user-sensitive or secure data.
- API rate limits must be respected to prevent account suspensions.

3.4.3 Security Requirements :

- Secure handling of API credentials and tokens.
- Use of environment variables to store sensitive information.

3.4.4 Software Quality Attributes :

- **Modularity**: Each system component (scraping, sentiment, forecasting) is separated for ease of maintenance.
- **Scalability**: New sentiment models or data sources can be added easily.
- **Reusability**: Components can be reused in other sentiment analysis pipelines.

3.5 System Requirements :

3.5.1 Database Requirements :

- Lightweight storage using CSV/JSON files for intermediate data.
- Optional use of SQLite or MongoDB for structured storage if needed.

3.5.2 Software Requirements :

- OS: Windows/Linux/macOS
- Python 3.9 or later
- Required libraries: BeautifulSoup, NLTK, SpaCy, Transformers, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib

3.5.3 Hardware Requirements :

- Minimum:
    - Processor: Dual-core 2.5 GHz
    - RAM: 8 GB
    - Storage: 500 MB free space

- Recommended:
    - Processor: Quad-core 3.0 GHz
    - RAM: 16 GB or more
    - GPU (for faster deep learning model inference): NVIDIA GTX 1060 or better

3.6 Analysis Models: SDLC Model to be Applied

We adopted the **Iterative Development Model** for this project.

- **Initial Phase**: Data collection pipeline design and basic scraper implementation.
- **Second Phase**: Sentiment models were tested and compared.
- **Third Phase**: Integration with forecasting logic and refinement using experimental results.
- **Final Phase**: Evaluation, analysis, and documentation.

This model allowed continuous feedback and improvement at each stage, making it ideal for a research-based system development like ours.

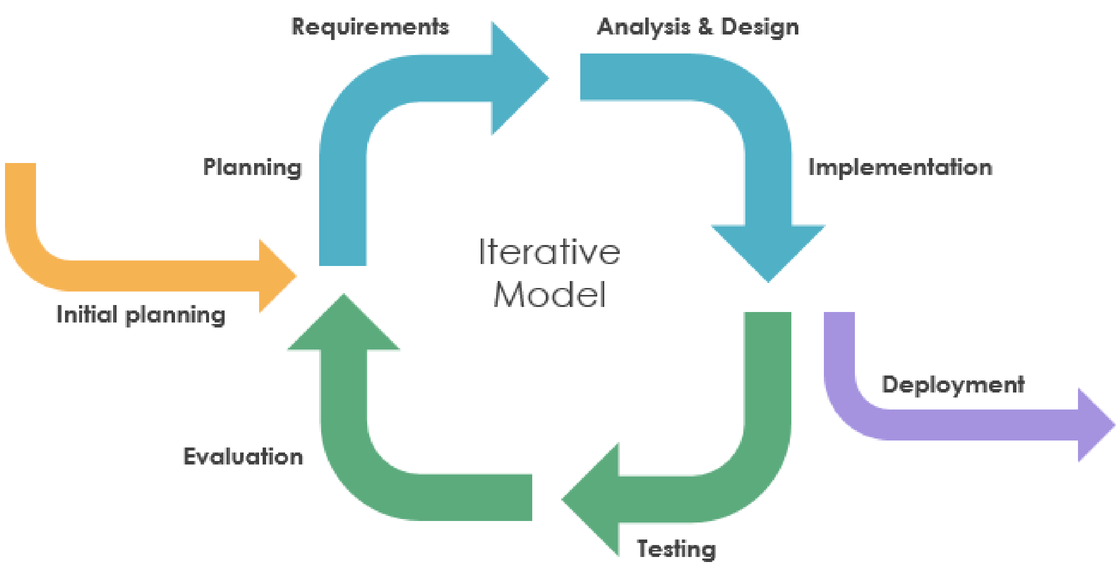

Fig. 1 : Iterative Development Model

CHAPTER 4
SYSTEM ARCHITECHTURE

4.1 System Architecture :

The system is structured into four main modules: Data Collection**,** Sentiment Analysis**,** Sentiment Aggregation**,** and Demand Forecasting**.** These modules communicate in a pipeline, where output from one stage is used as input for the next.

- **Data Collection Layer**: Responsible for scraping data from Amazon. It fetches product reviews using BeautifulSoup.
- **Sentiment Analysis Layer**: Applies multiple sentiment analysis models (VADER, TextBlob, DistilBERT, LSTM) to process and score the textual data.
- **Aggregation Layer**: Combines results from different sentiment models using a pooling function to create a robust sentiment profile.
- **Forecasting Layer**: Uses Bayesian Inference and decision tree-based models to predict demand based on sentiment, sales history, and seasonal patterns.

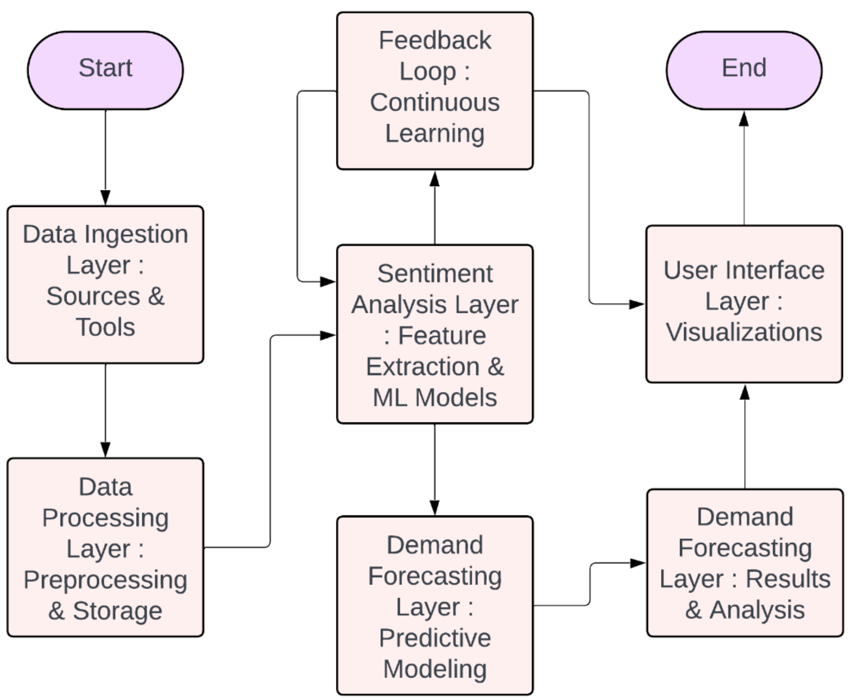

Fig. 2 : System Architectre

4.2 Mathematical Model :

The core mathematical model used in this project is a Bayesian Inference framework combined with ensemble learning to perform demand prediction. The key formulae are:

Sentiment Pooling Equation:

$$D= f(X_s,X_h,X_t) = 1/N \sum T_i(X)$$

Where:

- D = predicted demand
- X_s = sentiment-based features
- X_h = historical sales data
- X_t = time-based/seasonal features
- T_i= individual decision tree in ensemble
- N = number of trees

Bayesian Smoothing for Sentiment Adjustment:

Smoothed Probability = Total Observations + $k.\alpha$ / Observed Count + α

4.3 Data Flow Diagrams :

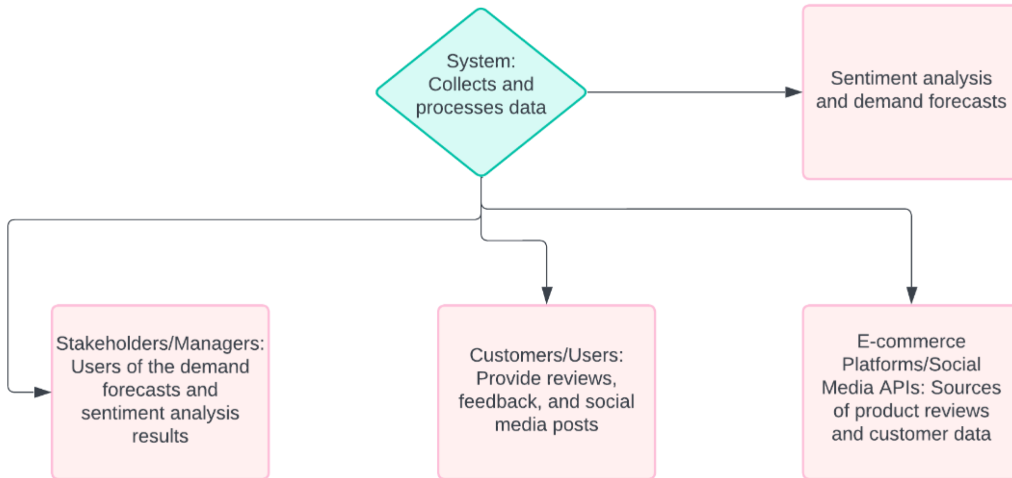

Fig. 3 : Data Flow Diagram

4.4 Entity Relationship Diagram :

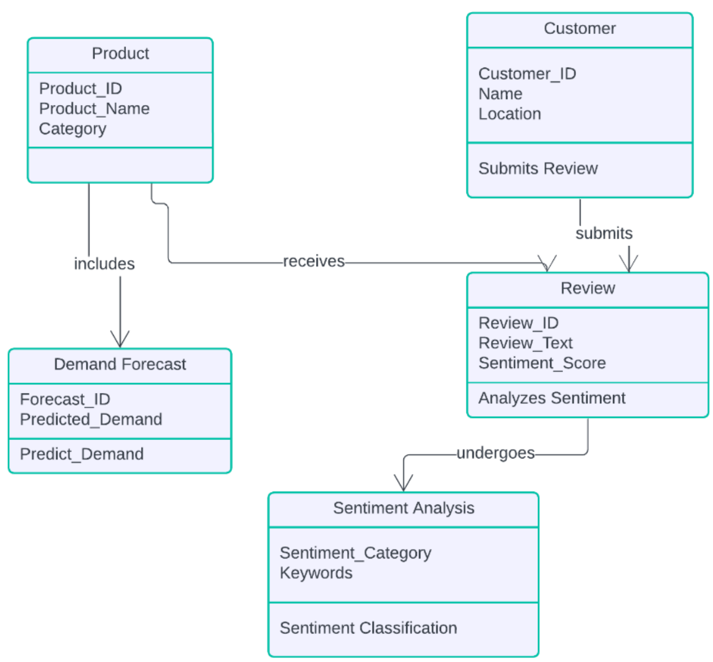

Fig 4 : ER Diagram

4.5 Activity Diagram :

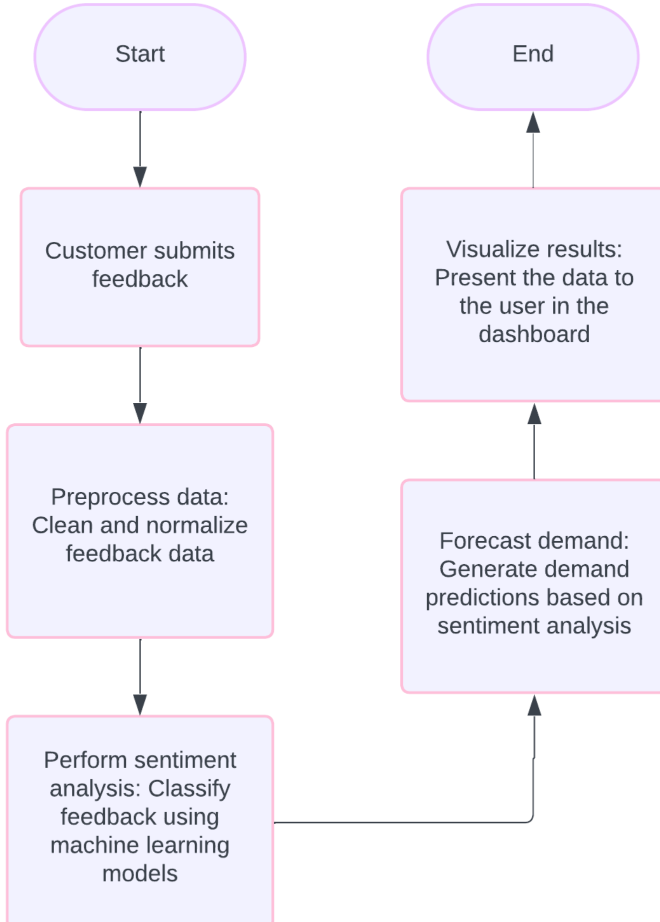

Fig. 5 : Activity Diagram

4.4 Class Diagram:

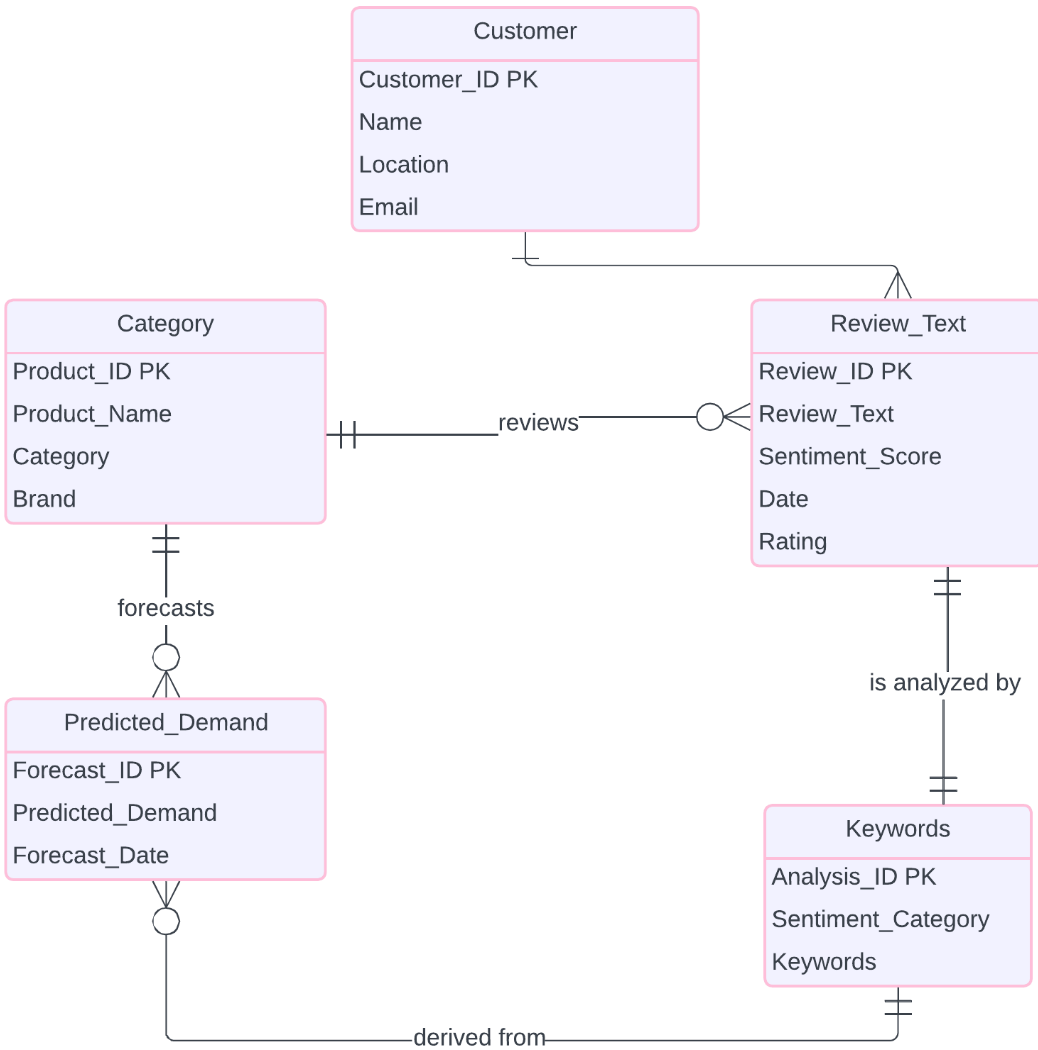

Fig. 6 : Class Diagram

CHAPTER 5
PROJECT PLAN

5.1 Project Estimates :

5.1.1 Reconciled Estimates :

The estimated effort and cost for the project were calculated based on the complexity of tasks involved in data scraping, sentiment analysis, model training, and demand prediction. We considered development time, testing, and documentation, with a moderate resource expenditure mainly focused on computational requirements and software tools.

5.1.2 Project Resources :

- **Software Tools**: Python, Jupyter Notebook, BeautifulSoup, Selenium, TensorFlow/Keras, HuggingFace Transformers, Matplotlib, Pandas.
- **Hardware**: Personal computers with minimum 8GB RAM and GPU support for deep learning.

5.2 Risk Management :

5.2.1 Risk Identification :

- Data scraping limitations due to website restrictions.
- Limited availability of labeled sentiment datasets.
- High resource consumption during model training.
- Possibility of incorrect sentiment classification due to domain-specific language.

5.2.2 Risk Analysis :

Each risk was assessed on the basis of its likelihood and potential impact. For example, model accuracy risk was rated high-impact, medium-likelihood; scraping limitations were rated medium-impact, high-likelihood.

5.2.3 Overview of Risk Mitigation, Monitoring, Management :

- Using multiple scraping techniques and backup datasets to avoid scraping failure.
- Regular validation of model outputs to prevent drift.
- Ensuring progress checkpoints are reviewed weekly by the team.
- If computational limitations arise, lightweight models or cloud resources will be used.

5.3 Team Organization :

5.3.1 Team Structure :

The project was executed by a team of three members, each of whom contributed equally across all stages of development. Tasks such as data collection, model implementation, evaluation, and documentation were collaboratively handled. Roles were rotated as needed to ensure balanced participation and collective learning.

5.3.2 Management Reporting and Communication :

Team members maintained regular coordination through weekly meetings and shared workspaces such as Google Drive and GitHub. All critical decisions and updates were mutually discussed and implemented, ensuring transparency and consistency throughout the project lifecycle. Periodic updates and reviews were also conducted with the faculty guide.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules :

The system is divided into multiple modules that work in tandem to enable sentiment-driven demand forecasting. Each module focuses on a critical aspect of the pipeline and contributes to the overall functioning of the system. The key modules include:

- **Data Scraping Module**: Responsible for extracting user-generated content (product reviews) from e-commerce and social media platforms.
- **Preprocessing Module**: Cleans and prepares raw textual data for sentiment analysis by removing noise, handling missing values, and performing tokenization.
- **Sentiment Analysis Module**: Uses a combination of pre-trained NLP models and a custom deep learning classifier to analyze sentiment.
- **Pooling & Aggregation Module**: Combines sentiment results from multiple models to generate a unified sentiment score.
- **Demand Forecasting Module**: Predicts future demand using sentiment scores, historical trends, and Bayesian inference techniques.

6.2 Tools and Technologies Used ::

- **Python**: Primary programming language used.
- **BeautifulSoup**: For scraping product reviews from Amazon.
- **NLTK and SpaCy**: For natural language processing tasks including tokenization, stopword removal, POS tagging, and named entity recognition.
- **Transformers (HuggingFace)**: For implementing BERT-based sentiment analysis.
- **Keras & TensorFlow**: For training and evaluating the custom deep learning sentiment model.
- **Matplotlib & Seaborn**: For data visualization and plotting results.
- **Scikit-learn**: For evaluation metrics and preprocessing utilities.
- **NumPy & Pandas**: For data manipulation and analysis.

6.3 Algorithm Details :

6.3.1 Sentiment Model Pooling Algorithm :

This algorithm aggregates predictions from multiple sentiment models (e.g., BERT, TextBlob, VADER, Custom LSTM) to improve classification robustness and reduce individual model bias. The formula for the ensemble prediction is:

$$D= f(X_s,X_h,X_t) = 1/N \sum T_i(X)$$

Where:

- D = predicted demand
- X_s = sentiment-based features
- X_h = historical sales data
- X_t = time-based/seasonal features
- T_i= individual decision tree in ensemble
- N = number of trees

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing :

To ensure the accuracy, reliability, and efficiency of our sentiment-driven demand forecasting system, various testing methodologies were applied during different stages of development:

- **Unit Testing**: Each module such as data scraping, preprocessing, sentiment analysis, and demand forecasting was tested individually to validate the correctness of functions and methods.
- **Integration Testing**: Modules were combined incrementally and tested to verify their interaction, especially between the sentiment analysis outputs and the demand forecasting module.
- **System Testing**: End-to-end testing of the complete pipeline was conducted to ensure that the system performs as intended under different data inputs.
- **Validation Testing**: The outputs of the sentiment analysis models and forecasting module were compared against expected outcomes to ensure the correctness of predictions.
- **Performance Testing**: Measured the time efficiency of scraping, analysis, and prediction tasks to ensure scalability for larger datasets.

7.2 Test Cases & Test Results :

- To ensure the robustness of our implementation, we defined and executed several key test cases aligned with the main functional blocks of the project:
- We began by verifying the web scraping functionality. The Amazon scraper correctly retrieved review texts, ratings, and timestamps when provided with a product URL.
- In the text preprocessing module, raw inputs such as "@user 😡 this phone sucks!!!" were correctly transformed into normalized text: "this phone sucks", with all irrelevant characters, emojis, and mentions removed.

- For sentiment analysis, each model was tested individually. VADER produced a compound score of 0.73 for the sentence “Amazing battery life and fast performance,” which was correctly interpreted as positive. Similarly, the DistilBERT model classified lengthy and nuanced reviews with high accuracy. The model pooling function was validated to ensure that outputs from VADER, TextBlob, DistilBERT, and LSTM were combined to produce a reliable consensus sentiment label using majority voting.
- Bayesian smoothing was tested by simulating a low-review region (e.g., a country with only 4 reviews), where the smoothed sentiment score was accurately adjusted using prior probabilities, avoiding overfitting to small sample sizes.
- In the demand distribution module, we provided sentiment proportions from multiple countries—such as 80% positive from India, 50% from the UK, and 90% from the USA. The module correctly normalized and allocated unit distributions across countries using the Bayesian-weighted sentiment scores and a greedy rounding technique.
- We also tested the LSTM-based sentiment classifier, trained on 3.6 million Amazon reviews. When fed test reviews like “Worst purchase I’ve ever made,” the model accurately predicted a negative label, with classification metrics achieving 82% accuracy overall.
- Finally, an end-to-end test confirmed the entire pipeline’s functionality. Given a product URL and relevant keywords, the system successfully performed scraping, analysis, and forecasting to output a country-wise CSV containing demand estimates.

CHAPTER 8
RESULTS

8.1 Outcomes :

This section presents the results obtained from the implementation of the sentiment-driven demand forecasting system. The evaluation was carried out in two main phases: sentiment classification and demand forecasting.

Sentiment Classification Performance :

The custom LSTM model was trained on a dataset of 3.6 million Amazon reviews and tested on a set of 2,000 labeled reviews. The classification performance is summarized below:

- **Negative Class**
  - Precision: 0.85
  - Recall: 0.76
  - F1-Score: 0.80
  - Support: 976 instances
- **Positive Class**
  - Precision: 0.79
  - Recall: 0.87
  - F1-Score: 0.83
  - Support: 1024 instances
- **Overall Accuracy**: 82%
- **Macro Average (Precision, Recall, F1-score)**: 0.82
- **Weighted Average (Precision, Recall, F1-score)**: 0.82

Sentiment Distribution by Region

The sentiment analysis model was further applied to reviews grouped by country. The results, visualized using a heatmap, revealed distinct regional sentiment patterns:

- **India** and **USA** showed predominantly positive sentiment.

- **UK** exhibited a more mixed sentiment distribution, indicating more balanced customer opinions.

This analysis is crucial in understanding market mood at a regional level.

Demand Forecasting Using Bayesian Smoothing

A sentiment-weighted allocation strategy was implemented using Bayesian inference to distribute a hypothetical stock of 1,000 devices among three countries. The resulting allocation based on sentiment scores was:

- **India**: 357 units
- **USA**: 357 units
- **UK**: 286 units

8.2 Screenshots :

Illustrates the convergence of the model over epochs and the balance between training and generalization.

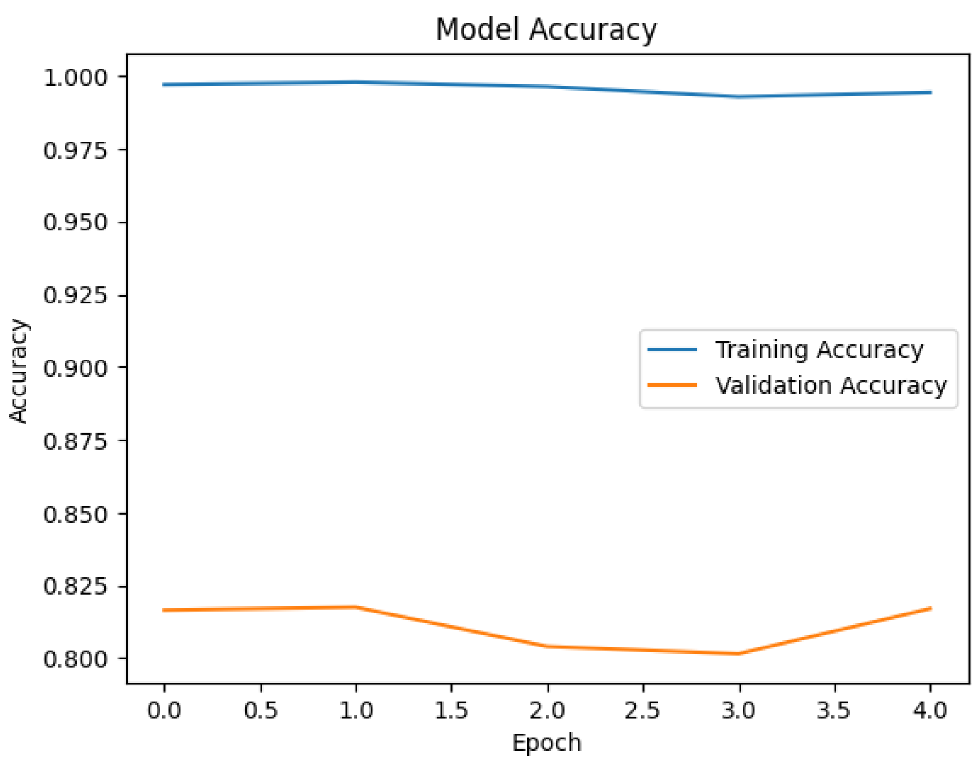

Fig. 7**:** Epoch Training Accuracy

Shows the distribution of true positives, true negatives, false positives, and false negatives for both sentiment classes.

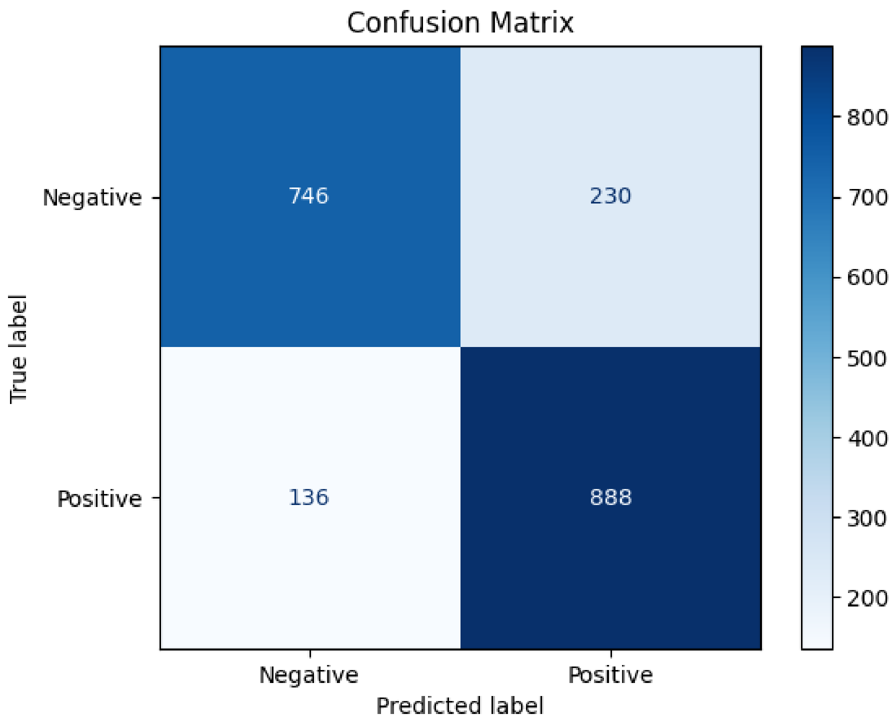

Fig 8: Confusion Matrix

Highlights regional differences in consumer sentiment extracted from product reviews.

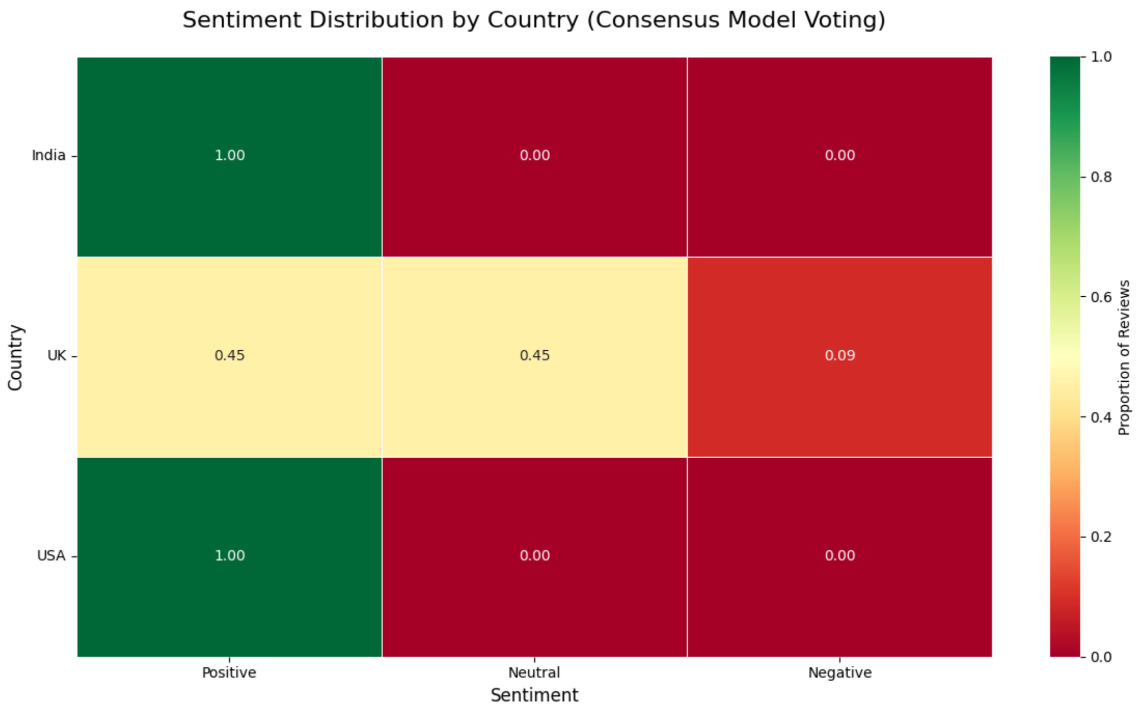

Fig 9: Heat Map of Sentiment Class

Displays the final distribution of 1,000 devices across India, USA, and UK based on calculated sentiment weights.

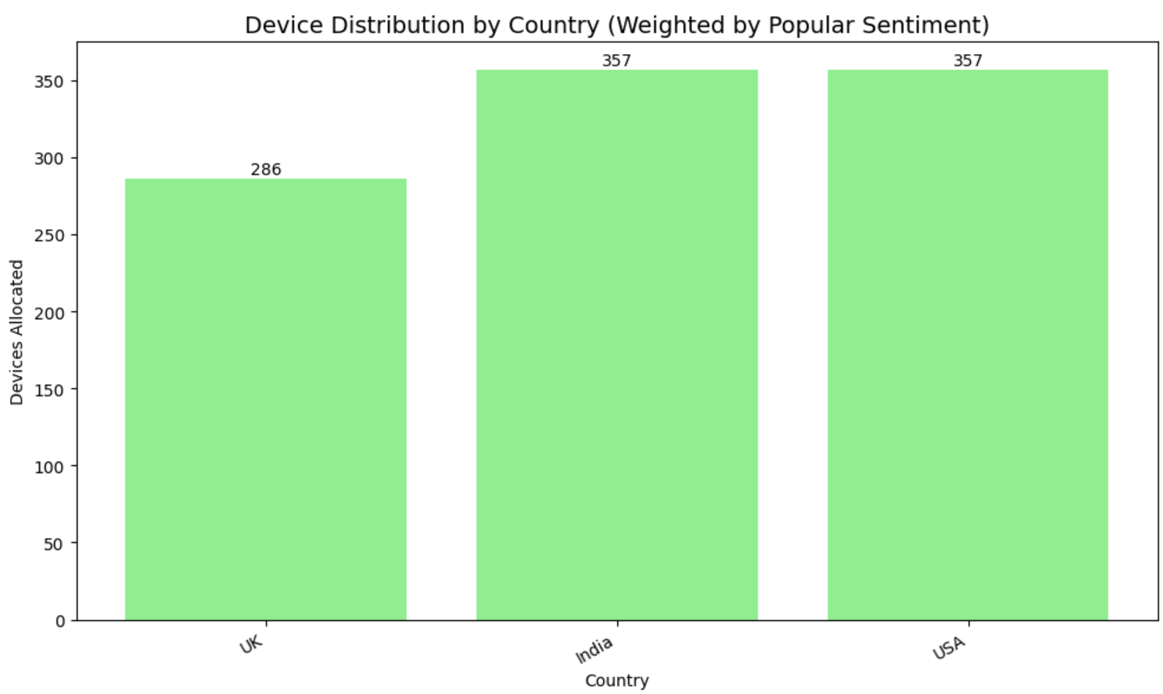

Fig. 10 : Device Distribution Based on Sentiment Scores (Bar Graph)

CHAPTER 9
CONCLUSION

9.1 Conclusions :

This project successfully demonstrates the implementation of a sentiment-driven demand forecasting system using a multi-model sentiment analysis approach integrated with a Bayesian inference model. By leveraging customer sentiment from platforms like Amazon, the system effectively interprets public opinion to influence strategic decisions such as regional product allocation.

The sentiment classification achieved an accuracy of **82%**, with strong performance metrics for both positive and negative classes. The pooling of models (VADER, TextBlob, BERT, and a custom LSTM) allowed the system to capture both contextual and emotional nuances in user reviews, resulting in more reliable predictions.

Furthermore, the Bayesian-smoothed sentiment weighting technique proved to be a robust method for fair and realistic demand distribution, especially when dealing with countries having imbalanced or sparse review data. The incorporation of regional sentiment enabled an adaptable, data-driven forecasting system that operates effectively even in rapidly changing market conditions.

9.2 Future Work :

- **Real-time deployment**: Future work may involve deploying the system as a real-time API or dashboard that continually updates demand predictions based on live sentiment data.
- **Expansion to more data sources**: Incorporating data from platforms like Reddit, YouTube comments, or Google Trends could enrich sentiment insights.
- **Multilingual Sentiment Analysis**: Incorporating sentiment detection in multiple languages could improve prediction accuracy for non-English reviews.
- **Advanced Forecasting Models**: Integrating deep learning-based forecasting methods like LSTMs or Prophet for time-series prediction may yield even more accurate future demand estimates.

9.3 Applications :

This project has the potential to be applied across multiple sectors, including:

- **Retail and E-commerce**: To better anticipate customer interest in products, enhancing promotional strategies.
- **Market Research and Trend Analysis**: For companies to gauge public opinion on products and adjust their strategies accordingly.
- **Product Launch Strategy**: To determine which regions show the highest excitement or interest based on online sentiment, guiding initial stock placement.
- **Sentiment-based Decision Support Systems**: Can be extended into business intelligence tools for better decision-making.

APPENDIX

Appendix A :

Problem Statement Feasibility Assessment :

The feasibility assessment of our project revolves around evaluating the complexity of sentiment analysis using machine learning and relevant mathematical models. Satisfiability analysis helps determine if a solution exists that satisfies all constraints in sentiment classification. This is analogous to solving a constraint satisfaction problem (CSP), where the constraints are based on the sentiment categories (positive, negative, neutral). For simpler models, these problems can be solved in polynomial time, categorized as P-type problems, ensuring feasibility for large datasets.

However, when incorporating more sophisticated models, feature optimization, or hyperparameter tuning, the complexity can escalate to NP-Hard or NP-Complete classifications. In such cases, exact solutions are computationally intensive. Techniques such as gradient descent or approximation algorithms are employed to find near-optimal solutions efficiently. While the problem's complexity can increase, careful model selection and optimization strategies maintain the project's overall feasibility.

Appendix B :

Plagiarism Report :

Appendix C :

Implementation Paper :

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